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Purposefully different,

consistently excellent

Welcome to EEN1037: Web Applications Development

HTTP & XHTML

Week 2

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Background Technologies

- What happens when we view a webpage?
- Hypertext Transfer Protocol (HTTP) – protocol between web browsers and servers
- Simple, information passed as plain text via TCP
- Telnet HTTP
  - Connect and send a HTTP Request
  - GET /index.html HTTP/1.0
  - Web Server returns requested page
  - If Browser used, page interpreted graphically
  - Can use HEAD instead of GET for response headers

```
$ telnet www.google.com 80
Connected to www.google.com.
GET /index.html HTTP/1.1

HTTP/1.1 200 OK
Expires: -1
Cache-Control: private, max-age=0
Content-Type: text/html; charset=ISO-8859-1
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Accept-Ranges: none
Vary: Accept-Encoding
Transfer-Encoding: chunked

8000
<!doctype html><html itemscope=""
itemtype="http://schema.org/WebPage" lang="en-IE">
<head><meta content="text/html; charset=UTF-8" http-
equiv="Content-Type"><meta
content="/images/branding/googleg/1x/googleg_standard_color_1
28dp.png" itemprop="image"><title>Google</title>
...
```

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HTTP Version 1.0

- HTTP 1.0 is simple and stateless protocol
- Client makes a request, server responds -> quick simple transaction
- Connection made for every object to be downloaded
- Wasteful of resources – packet overhead involved in getting pages with many separate parts

-> HTTP Version 1.1

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HTTP Version 1.1

- HTTP/1.1 defined by W3C to address a number of issues with HTTP/1.0
- Basic operation remains the same and clients/servers are backwards compatible
- HTTP/1.1 defines persistent connections
- Numerous documents downloaded over one connection
- HTTP/1.1 persistent connections default unless otherwise specified

```
HEAD /index.html HTTP/1.1
Host: www.eeng.dcu.ie
```

- New features on proxies and caching

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Method Types

- Clients send requests as one of a number of different types

- **GET** Method – used for retrieving information from the server. Can include additional information, describing what it wishes to obtain

  https://loop.dcu.ie/course/view.php?id=51621

- **POST** Method – uses different technique for transferring data. Used for sending large quantities of information, for which GET is unsuitable URL remains the same always, not bookmarkable/emailed/reloaded

  http://www.server.com/scripts/dologin

- **HEAD** Method – acts like a GET with a blank page, sends response headers

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Method Types

- Remaining method types not commonly used
- **PUT** Method – used for placing documents on the server
- **DELETE** Method – used for deleting documents from the server
- **TRACE** Method – used for debugging (returns request contents)
- **OPTIONS** Method – used to ask server which methods it supports etc.

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Hypertext Markup Language

- What are HTML documents?
- URLs (Universal Resource Locator):
  http://www.server.com/path/file.html
- Development using WYSIWYG (What you see is what you get) tools
- Uploading to the Web Server (FTP, Samba, Local etc.)
- Basic Example
- Writing HTML manually – the proper way!
- `<HTML>, <HEAD>, <TITLE>, <BODY>, <BR>, <P>, <H1>, <FONT>, <B>, <I>, <A>, <UL>, <LI>, <PRE>, <IMG>, <TABLE>, <TR>, <TD>`
- Attributes
- World Wide Web Consortium (W3C)

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HTML (seen in Week 1)

- HyperText Markup Language

```
<!DOCTYPE html>
<html>
    <body>
        <h1>My First Heading</h1>
        <p>My first paragraph.</p>
    </body>
</html>
```

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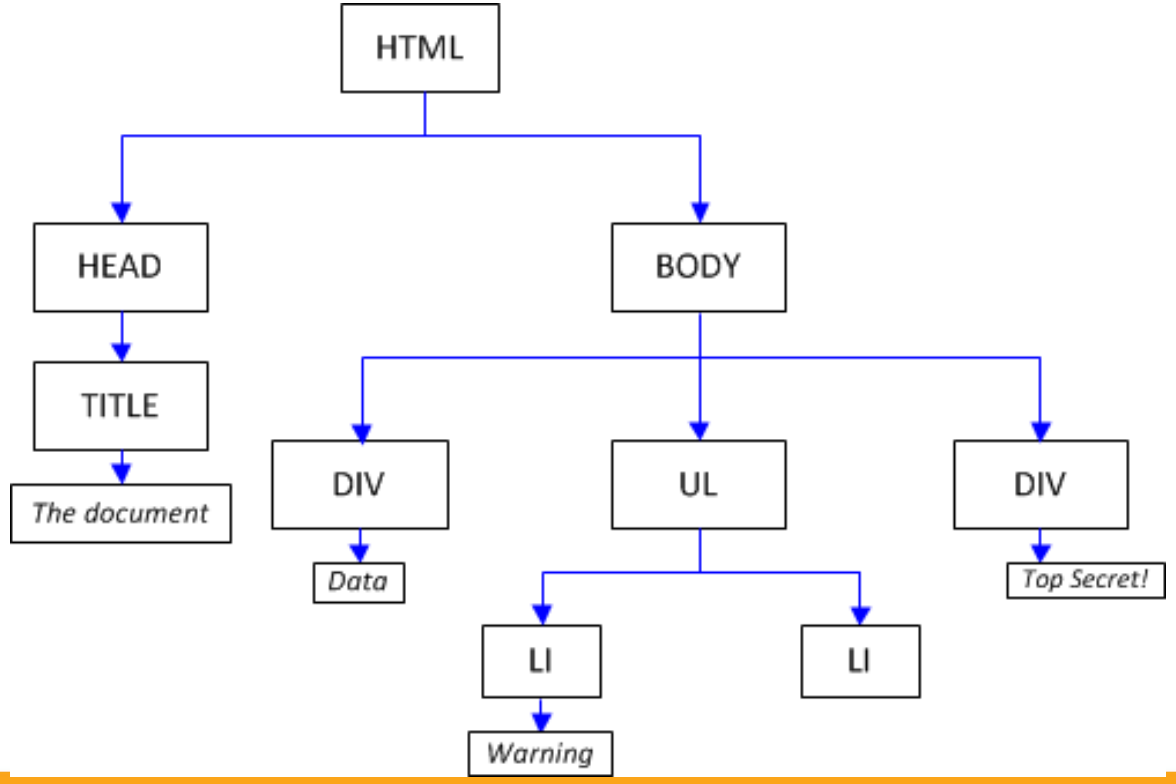
HTML DOM (seen in Week 1)

- Defines a standard for accessing and manipulating HTML documents



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XML

- eXtensible Markup Language
- “generalization” of HTML
- A generic exchange format not only for rendering information in browsers…
  - … but for arbitrary applications that support XML

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XML Example

```
<FILM>
 <TITLE> Titanic </TITLE>

 <PRODUCER>
       <NAME>James Cameron</NAME>
       <NAME>Jon Landau</NAME>
</PRODUCER>

 <DIRECTOR>James Cameron</DIRECTOR>

  <SCREENWRITER>James Cameron</SCREENWRITER>

 <DISTRIBUTOR>Paramount</DISTRIBUTOR>

 <BOX-OFFICE>$376,270,721</BOX-OFFICE>
</FILM>
```

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HTML vs XHTML

- XHTML is a reformulation of HTML into an XML format

- Features

  - Much stricter and less forgiving format
  - Case sensitive on all tags and attributes
  - XHTML must be well formed

- Number of advantages for us when working with JavaScript, Ajax, DOM, JQuery etc.

- We will use XHTML type tags in module

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XHTML

- XHTML = XML + HTML
- HTML is loosely structured,
  - can work even if some tags are not closed
  - Invalid tags nesting/overlap
- Addition of XML tags to make validatable XML
- XHTML follows strictly well-formedness rules
- W3C XHTML Validator
- Mostly when we will write XHTML we will ignore first lines for simplicity
- W3Schools TryIT HTML Editor

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XHTML/HTML Accessibility

- Designing web pages so as to make websites users to individuals of all abilities and disabilities
- “alt” attribute in <img> elements for example
- WCAG (Web Content Accessibility Guidelines) 1.0 Published in 1999 by the W3C. Version 2.0 in 2008
- Ethical & Moral Responsibility .... Legal responsibility?
- Disability Act 2005 in Ireland
- "Where a public body communicates in electronic form with one or more persons, the head of the body shall ensure, that as far as practicable, the contents of the communication are accessible to persons with a visual impairment to whom adaptive technology is available.“
- Web Accessibility Checkers – common sense also needed!

HTML Forms

- Client front-end to our server-side applications
- Exist on the client side, interpreted within the browser
- Typically require a server-side application to make them useful
- HTML forms contain normal content, markup and special elements called controls and their labels
- Control's "control name" is given by its name attribute
- `<FORM METHOD="POST" ACTION="http://…../formhandler" NAME="myform">`
- `<INPUT TYPE="submit" VALUE="Submit Query">`

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HTML Form

First Name :

Last Name :

Date of Birth :

Email id :

Mobile Number :

SUBMIT RESET

tutorialBrain.com

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Document Object Model (DOM)

- Allows us to programmatically access and manipulate a web page or XML page
- Provides an object-oriented representation of the elements and content of a page combined with methods for setting and retrieving elements and data
- DOM provides us with an API (Application Programming Interface) we can use to do this
- Standardised by the W3C
  - DOM1 Recommendation 1998
  - DOM2 Recommendation 2000
  - DOM3 Recommendation 2004
  - Supported in all modern browsers

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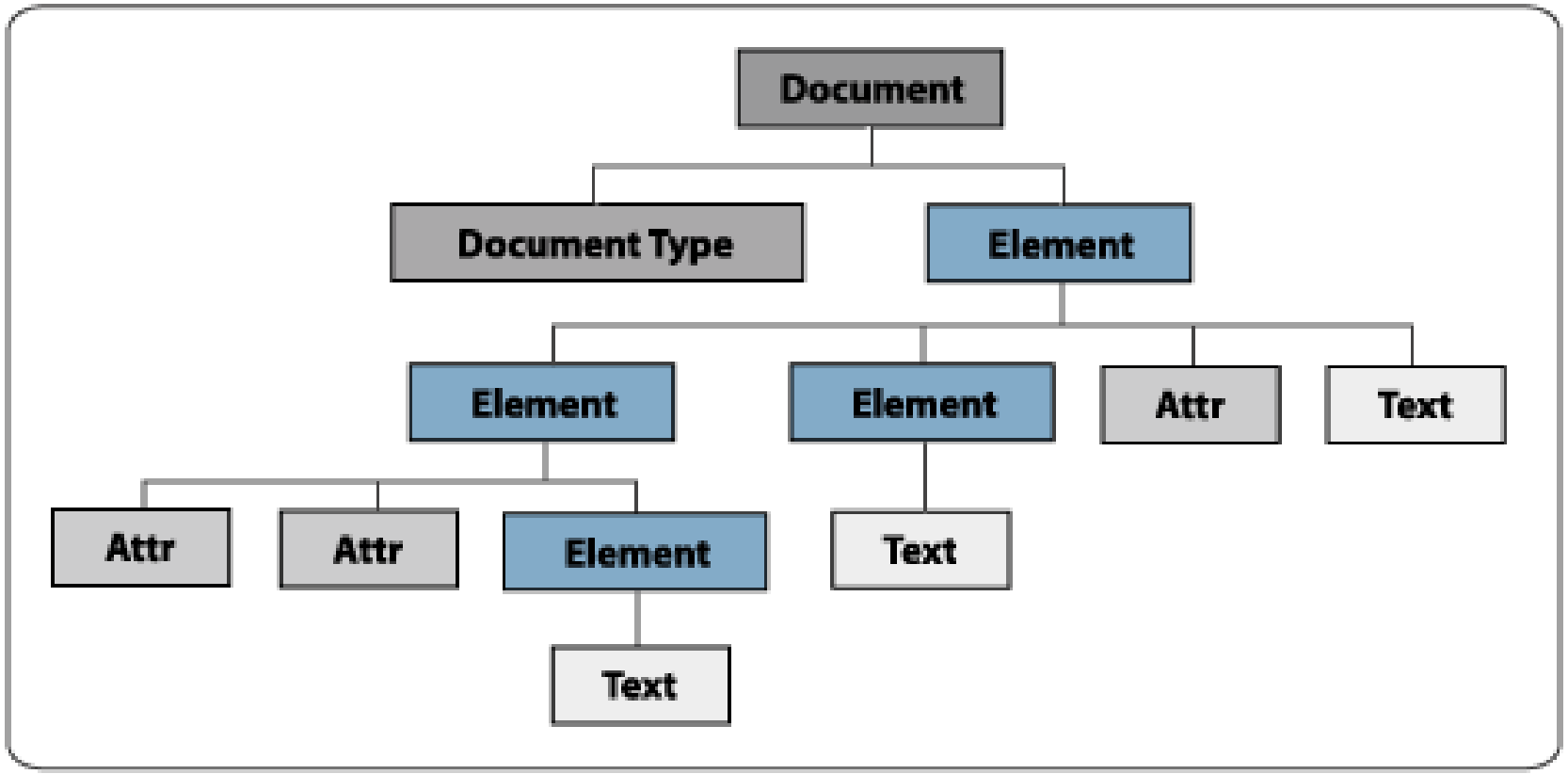
DOM Document Tree



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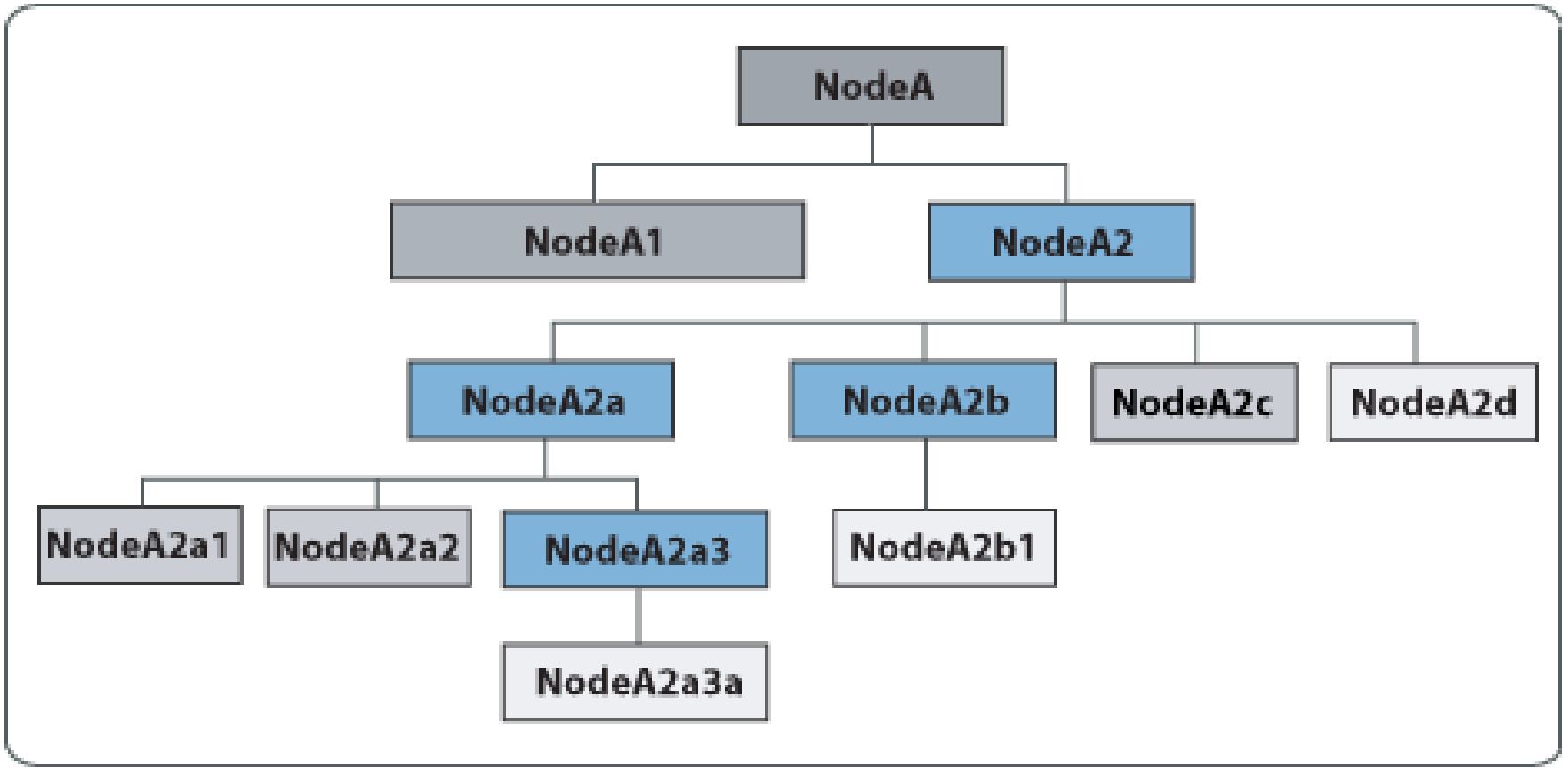
DOM Document Tree

- **All content and elements represented as nodes**



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